are yet further Answer Schemas, such as ω^3;, ω^4, ω^5, and so on. It goes on quite a ways, you know.

Achilles: I can imagine, I suppose it comes to Answer Schema are yet further Answer Schemas, such as w;, w4, w5, and so on. It goes on quite a ways, you know.

Achilles: I can imagine, I suppose it comes to Answer Schema ω^{ω} after a while.

Tortoise: Of course.

Achilles: And then $\omega^{\omega\omega}$, and $\omega^{\omega\omega\omega}$',

Tortoise: You're catching on mighty fast, Achilles. I have a suggestion for you, if you don't mind. Why don't you throw all of those together into a single Answer Schema?

Achilles: All right, though I'm beginning to doubt whether it will do any good.

Tortoise: It seems to me that within our naming conventions as so far set up, there is no obvious name for this one. So perhaps we should just arbitrarily name it Answer Schema Œo.

Achilles: Confound it all! Every time you give one of my answers a NAME, it seems to signal the imminent shattering of my hopes that that answer will satisfy you. Why don't we just leave this Answer Schema nameless?

Tortoise: We can hardly do that, Achilles. We wouldn't have any way to refer to it without a name. And besides, there is something inevitable and rather beautiful about this particular Answer Schema. It would be quite ungraceful to leave it nameless! And you wouldn't want to do something lacking in grace on your birthday, would you? Or is it your birthday? Say, speaking of birthdays, today is MY' birthday!

Achilles: It is?

Tortoise: Yes, it is. Well, actually, it's my uncle's birthday, but that's almost the same. How would you like to treat me to a delicious birthday dinner this evening?

Achilles: Now just a cotton-picking minute, Mr. T. Today is MY birthday. You should do the treating!

Tortoise: Ah, but you never did succeed in convincing me of the veracity of that remark. You kept on beating around the bush with answers, Answer Schemas, and whatnot. All I wanted to know was if it was your birthday or not, but you managed to befuddle me entirely. Oh, well, too bad. In any case, I'll be happy to let you treat me to a birthday dinner this evening.

Achilles: Very well. I know just the place. They have a variety of delicious soups. And I know exactly what kind we should have ...

CHAPTER XV

Jumping out of the System

A More Powerful Formal System

ONE OF THE things which a thoughtful critic of Gödel's proof might do would be to examine its generality. Such a critic might, for example, suspect that Gödel has just cleverly taken advantage of a hidden defect in one particular formal system, TNT. If this were the case, then perhaps a formal system superior to **TNT** could be developed which would not be subject to the Gödelian trick, and Gödel's Theorem would lose much of its sting. In this Chapter we will carefully scrutinize the properties of **TNT** which made it vulnerable to the arguments of last Chapter.

A natural thought is this: If the basic trouble with **TNT** is that it contains a "hole"-in other words, a sentence which is undecidable, namely **G**-then why not simply plug up the hole? Why not just tack **G** onto **TNT** as a sixth axiom? Of course, by comparison to the other axioms, **G** is a ridiculously huge giant, and the resulting system-**TNT+G**-would have a rather comical aspect due to the disproportionateness of its axioms. Be that as it may, adding **G** is a reasonable suggestion. Let us consider it done. Now, it is to be hoped, the new system, **TNT+G**, is a superior formal system-one which is not only supernatural-free, but also complete. It is certain that **TNT+G** is superior to **TNT** in at least one respect: the string **G** is no longer undecidable in this new system, since it is a theorem.

What was the vulnerability of **TNT** due to? The essence of its vulnerability was that it was capable of expressing statements about itself-in particular, the statement

"I Cannot Be Proven in Formal System **TNT**"

or, expanded a bit,

> "There does not exist a natural number which forms a **TNT**-proof-pair with the Gödel number of this string."

Is there any reason to expect or hope that **TNT+G** would be invulnerable to Gödel's proof? Not really. Our new system is just as expressive as **TNT**. Since Gödel's proof relies primarily on the expressive power of a formal system, we should not be surprised to see our new system succumb,

too. The trick will be to find a string which expresses the statement

"I Cannot Be Proven in Formal System **TNT+G**."

Actually, it is not much of a trick, once you have seen it done for **TNT**. All the same principles are employed: only, the context shifts slightly. (Figuratively speaking, we take a tune we know and simply sing it again, only in a higher key.) As before, the string which we are looking for-let us call it "**G**'"-is constructed by the intermediary of an "uncle", But instead of being based on the formula which represents **TNT**-proof-pairs, it is based on the similar but slightly more complicated notion of **TNT+G**-proofpairs. This notion of **TNT+G**-proof-pairs is only a slight extension of the original notion of' **TNT**-proof-pairs.

A similar extension could be envisaged for the **MIU**-system. We have seen the unadulterated form of **MIU**-proof-pairs. Were we now to add MU as a second axiom, we would be dealing with a new system-the **MIU+MU** system. A derivation in this extended system is presented:

MU	axiom
MUU	rule 2

There is a **MIU+MU**-proof-pair which corresponds-namely, m = 30300, n = 300. Of course, this pair of numbers does not form a **MIU**-proof-pair-only a **MIU+MU**-proof-pair. The addition of an extra axiom does not substantially complicate the arithmetical properties of proof-pairs. The significant fact about them-that being a proof-pair is primitive recursive-is preserved.

The Gödel Method Reapplied

Vow, returning to **TNT+G**, we will find a similar situation**. TNT+G** proof-pairs, like their predecessors, are primitive recursive, so they are represented inside **TNT+G** by a formula which we abbreviate in an obvious manner.

(TNT+G)-PROOF-PAIR{a,a' }

Vow we just do everything all over again. We make the counterpart of **G** by beginning with an "uncle", just as before:

3a:3a':<(TNT+G)-PROOF-PAIR{a,a'}ARITHMOQUINE{a",a'}>

.et us say its Gödel-number is u'. Now we arithmoquine this very uncle. That will give us **G**':

3a:3a': < (TNT+G)-PROOF-PAIR{a,a' }
ARITHMOQUINE{SSS….SSSo/a´´,a´}>
U´ S´s

Its interpretation is

"There is no number *a* that forms a **TNT +G**-proof-pair
with the arithmoquinification of *u*'."

More concisely,

"I Cannot Be Proven in Formal System **TNT+G**."

Multifurcation

Well (yawn), the details are quite boring from here on out. G' is precisely to **TNT+G** as **G** was to **TNT** Itself. One finds that either **G**' or **-G**' can be added to **TNT+G**, to yield a further splitting of number theory. And, lest you think this only happens to the "good guys", this very same dastardly trick can be played upon **TNT+--G**-that is, upon the nonstandard extension of **TNT** gotten by adding **G**'s negation. So now we see (Fig. 75) that there are all sorts of bifurcations in number theory:

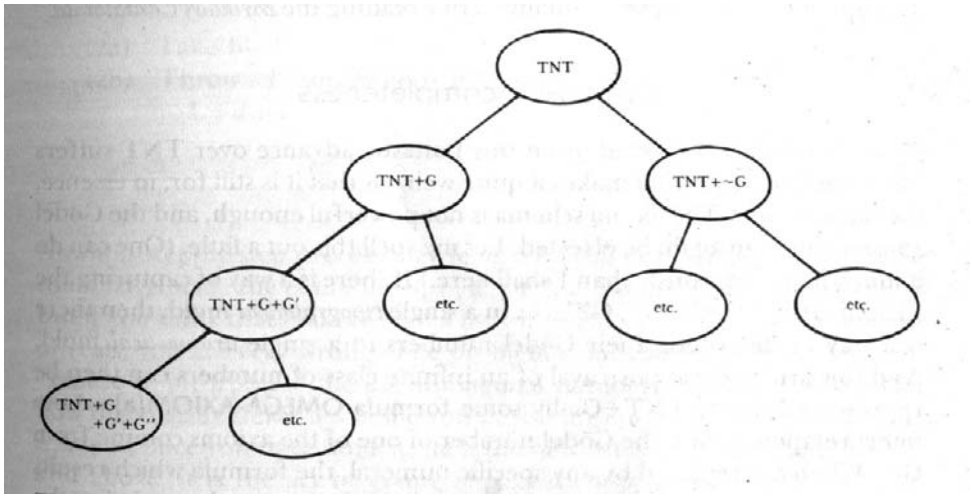


*FIGURE 75. "Multifurcation" of **TNT**. Each extension of **TNT** has its very own Gödel sentence; that sentence, or its negation, can be added on, so that from each extension there sprouts a pair of further extensions, a process which goes on ad infinitum.*

Of course, this is just the beginning. Let us imagine moving down the leftmost branch of this downwards-pointing tree, where we always toss in the Gödel sentences (rather than their negations). This is the best we can do by way of eliminating supernaturals. After adding **G**, we add **G**'. Then we add **G**", and **G**"', and so on. Each time we make a new extension of **TNT**, its vulnerability to the Tortoise's method-pardon me, I mean Gödel's method.. allows a new string to be devised, having the interpretation.

"I cannot be proven in formal system X"

Naturally, after a while, the whole process begins to seem utterly predictable and routine. Why, all the "holes" are made by one single technique! This means that, viewed as typographical objects, they are all cast from one single mold, which in turn means that one single axiom schema suffices to represent all of them! So if this is so, why not plug up all :he holes at once and be done with this nasty business of incompleteness 3nce and for all? This would be accomplished by adding an axiom schema to **TNT**, instead of just one axiom at a time. Specifically, this axiom schema would be the mold in which all of **G, G', G", G'"**, etc., are cast. By adding :his axiom schema (let's call it "**G~**"), we would be outsmarting the "Gödelization" method. Indeed, it seems quite clear that adding **G**. to **TNT** would :)e the last step necessary for the complete axiomatization of all of number-theoretical truth.

It was at about this point in the Contracrostipunctus that the Tortoise related the Crab's invention of "Record Player Omega". However, readers were left dangling as to the fate of that device, since before completing his tale, the tuckered-out Tortoise decided that he had best go home to sleep; but not before tossing off a sly reference to Gödel's Incompleteness Theorem). Now, at last, we can get around to clearing up that dangling detail ... Perhaps you already have an inkling, after reading the Birthday Cantatatata.

Essential Incompleteness

As you probably suspected, even this fantastic advance over **TNT** suffers the same fate. And what makes it quite weird is that it is still for, in essence, the same reason. The axiom schema is not powerful enough, and the Gödel construction can again be effected. Let me spell this out a little. (One can do it much more rigorously than I shall here.) If there is a way of capturing the various strings **G, G', G", G'"** . . in a single typographical mold, then there is a way of describing their Gödel numbers in a single arithmetical mold. And this arithmetical portrayal of an infinite class of numbers can then be represented inside **TNT+G**. by some formula **OMEGA-AXIOM**{a} whose interpretation is: "a is the Godel number of one of the axioms coming from **G**.". When a is replaced by any specific numeral, the formula which results will be a theorem of **TNT+G**. if and only if the numeral stands for the Gödel number of an axiom coming from the schema.

With the aid of this new formula, it becomes possible to represent even such a complicated notion as **TNT+G**.-proof-pairs inside **TNT+Gω**:

(TNT+G.)- PROOF- PAIR{a, a')

sing this formula, we can construct a new uncle, which we proceed to Arithmoquine in the by now thoroughly familiar way, making yet another undecidable string, which will be called "**TNT+Gω+i**". At this point, you might well wonder, "Why isn't **Gω+i** among the axioms created by the axiom schema **Gω**?" The answer is that **G** was not clever enough to foresee its own embeddability inside number theory.

In the Contracrostipunctus, one of the essential steps in the Tortoise's making an "unplayable record" was to get a hold of a manufacturer's blueprint of the record player which he was out to destroy. This was necessary so that he could figure out to what kinds of vibrations it was vulnerable, and then incorporate into his record such grooves as would code for sounds which would induce those vibrations. It is a close analogue to the Gödel trick, in which the system's own properties are reflected inside the notion of proof-pairs, and then used against it. Any system, no matter how complex or tricky it is, can be Gödel-numbered, and then the notion of its proof-pairs can be defined-and this is the petard by which it is hoist. Once a system is well-defined, or "boxed", it becomes vulnerable.

This principle is excellently illustrated by the Cantor diagonal trick, which finds an omitted real number for each well-defined list of reals between 0 and 1. It is the act of giving an explicit list-a "box" of reals which causes the downfall. Let us see how the Cantor trick can be repeated over and over again. Consider what happens if, starting with some list L, you do the following:

(la) Take list L, and construct its diagonal number d.
(lb) Throw d somewhere into list L, making a new list L+d.

(2a) Take list L +d, and construct its diagonal number d'.
(2b) Throw d' somewhere into list L+d, making a new list L+d+d'.

Now this step-by-step process may seem a doltish way to patch up **L**, for we could have made the entire list *d, d', d", d"'*, ... at once, given **L** originally. But if you think that making such a list will enable you to complete your list of reals, you are very wrong. The problem comes at the moment you ask, "Where to incorporate the list of diagonal numbers inside **L**?" No matter how diabolically clever a scheme you devise for ensconcing the d-numbers inside L, once you have done it, then the new list is still vulnerable. As was said above, it is the act of giving an explicit list-a "box" of reals-that causes the downfall.

Now in the case of formal systems, it is the act of giving an explicit recipe for what supposedly characterizes number-theoretical truth that causes the incompleteness. This is the crux of the problem with **TNT+Gω**,. Once you insert all the **G**'s in a well-defined way into TNT, there is seen to be some other G-some unforeseen **G**-which you didn't capture in your axiom schema. And in the case of the **TC**-battle inside the *ContracrostiPunctus*, the instant a record player's "architecture" is determined, the record player becomes capable of being shaken to pieces.

So what is to be done? There is no end in sight. It appears that **TNT**, even when extended ad infinitum, cannot be made complete. **TNT** is therefore said to suffer from *essential incompleteness* because the income-

pleteness here is part and parcel of **TNT**; it is an essential part of the nature of **TNT** and cannot be eradicated in any way, whether simpleminded or ingenious. What's more, this problem will haunt any formal version of number theory, whether it is an extension of **TNT**, a modification of **TNT**, or an alternative to **TNT**. The fact of the matter is this: the possibility of constructing, in a given system, an undecidable string via Gödel's self-reference method, depends on three basic conditions:

(1) That the system should be rich enough so that all desired statements about numbers, whether true or false, can be expressed in it. (Failure on this count means that the system is from the very start too weak to be counted as a rival to **TNT**, because it can't even express number-theoretical notions that **TNT** can. In the metaphor of the Contracrosttpunctus, it is as if one did not have a phonograph but a refrigerator or some other kind of object.)
(2) That all general recursive relations should be represented by formulas in the system. (Failure on this count means the system fails to capture in a theorem some general recursive truth, which can only be considered a pathetic bellyflop if it is attempting to produce all of number theory's truths. In the Contracrostipunctus metaphor, this is like having a record player, but one of low fidelity.)
(3) That the axioms and typographical patterns defined by its rules be recognizable by some terminating decision procedure. (Failure on this count means that there is no method to distinguish valid derivations in the system from invalid ones-thus that the "formal system" is not formal after all, and in fact is not even well-defined. In the Contracrostipunctus metaphor, it is a phonograph which is still on the drawing board, only partially designed.)

Satisfaction of these three conditions guarantees that any consistent system will be incomplete, because Gödel's construction is applicable.

The fascinating thing is that any such system digs its own hole; the system's own richness brings about its own downfall. The downfall occurs essentially because the system is powerful enough to have self-referential sentences. In physics, the notion exists of a "critical mass" of a fissionable substance, such as uranium. A solid lump of the substance will just sit there, if its mass is less than critical. But beyond the critical mass, such a lump will undergo a chain reaction, and blow up. It seems that with formal systems there is an analogous critical point. Below that point, a system is "harmless" and does not even approach defining arithmetical truth formally; but beyond the critical point, the system suddenly attains the capacity for self-reference, and thereby dooms itself to incompleteness. The threshold seems to be roughly when a system attains the three properties listed above.

Once this ability for self-reference is attained, the system has a hole which is tailor-made for itself; the hole takes the features of the system into account and uses them against the system.

The Passion According to Lucas

The baffling repeatability of the Gödel argument has been used by various people-notably J. R. Lucas-as ammunition in the battle to show that there is some elusive and ineffable quality to human intelligence, which makes it unattainable by "mechanical automata"-that is, computers. Lucas begins his article "Minds, Machines, and Gödel" with these words:

> Gödel's theorem seems to me to prove that Mechanism is false, that is, that minds cannot be explained as machines.'

Then he proceeds to give an argument which, paraphrased, runs like this. For a computer to be considered as intelligent as a person is, it must be able to do every intellectual task which a person can do. Now Lucas claims that no computer can do "Gödelization" (one of his amusingly irreverent terms) in the manner that people can. Why not? Well, think of any particular formal system, such as **TNT**, or **TNT+G**, or even **TNT+G**.. One can write a computer program rather easily which will systematically generate theorems of that system, and in such a manner that eventually, any preselected theorem will be printed out. That is, the theorem-generating program won't skip any portion of the "space" of all theorems. Such a program would be composed of two major parts: (1) a subroutine which stamps out axioms, given the "molds" of the axiom schemas (if there are any), and (2) a subroutine which takes known theorems (including axioms, of course) and applies rules of inference to produce new theorems. The program would alternate between running first one of these subroutines, and then the other.

We can anthropomorphically say that this program "knows" some facts of number theory-namely, it knows those facts which it prints out. If it fails to print out some true fact of number theory, then of course it doesn't "know" that fact. Therefore, a computer program will be inferior to human beings if it can be shown that humans know something which the program cannot know. Now here is where Lucas starts rolling. He says that we humans can always do the Gödel trick on any formal system as powerful as TNT-and hence no matter what the formal system, we know more than it does. Now this may only sound like an argument about formal systems, but it can also be slightly modified so that it becomes, seemingly, an invincible argument against the possibility of Artificial Intelligence ever reproducing the human level of intelligence. Here is the gist of it:

> Rigid internal codes entirely rule computers and robots; ergo ...
> Computers are isomorphic to formal systems. Now .. .
> Any computer which wants to be as smart as we are has got to be able to do number theory as well as we can, so….

Among other things, it has to be able to do primitive recursive arithmetic. But for this very reason .. .

It is vulnerable to the Gödelian "hook", which implies that ...

We, with our human intelligence, can concoct a certain statement of number theory which is true, but the computer is blind to that statement's truth (i.e., will never print it out), precisely because of Gödel's boomeranging argument.

This implies that there is one thing which computers just cannot be programmed to do, but which we can do. So we are smarter.

Let us enjoy, with Lucas, a transient moment of anthropocentric glory:

> However complicated a machine we construct, it will, if it is a machine, correspond to a formal system, which in turn will be liable to the Gödel procedure for finding a formula unprovable-in-that-system. This formula the machine will be unable to produce as being true, although a mind can see it is true. And so the machine will still not be an adequate model of the mind. We are trying to produce a model of the mind which is mechanical-which is essentially "dead"-but the mind, being in fact "alive," can always go one better than any formal, ossified, dead system can. Thanks to Gödel's theorem. the mind always has the last word.[2]

On first sight, and perhaps even on careful analysis, Lucas' argument appears compelling. It usually evokes rather polarized reactions. Some ;eize onto it as a nearly religious proof of the existence of souls, while others laugh it off as being unworthy of comment. I feel it is wrong, but Fascinatingly so-and therefore quite worthwhile taking the time to rebut. In fact, it was one of the major early forces driving me to think over the matters in this book. I shall try to rebut it in one way in this Chapter, and in ether ways in Chapter XVII.

We must try to understand more deeply why Lucas says the computer cannot be programmed to "know" as much as we do. Basically the idea is :hat we are always outside the system, and from out there we can always perform the "Gödelizing" operation, which yields something which the program, from within, can't see is true. But why can't the "Gödelizing operator", as Lucas calls it, be programmed and added to the program as a third major component, Lucas explains:

> The procedure whereby the Gödelian formula is constructed is a standard procedure-only so could we be sure that a Gödelian formula can be constructed for every formal system. But if it is a standard procedure, then a machine should be able to be programmed to carry it out too.... This would correspond to having a system with an additional rule of inference which allowed one to add, as a theorem, the Gödelian formula of the rest of' the formal system, and then the Gödelian formula of this new, strengthened, formal system, and so on. It would be tantamount to adding to the original formal system an infinite sequence of axioms, each the Gödelian formula of the system hitherto obtained… We might expect a mind, faced with a machine that possessed a Gödelizing operator, to take this into account, and

out-Gödel the new machine, Gödelizing operator and all. This has, in fact, proved to be the case. Even if we adjoin to a formal system the infinite set of axioms consisting of the successive Gödelian formulae, the resulting system is still incomplete, and contains a formula which cannot be proved-in-the system, although a rational being can, standing outside the system, see that it is true. We had expected this, for even if an infinite set of axioms were added, they would have to be specified by some finite rule or specification, and this further rule or specification could then be taken into account by a mind considering the enlarged formal system. In a sense, just because the mind has the last word, it can always pick a hole in any formal system presented to it as a model of its own workings. The mechanical model must be, in some sense, finite and definite: and then the mind can always go one better.'

Jumping Up a Dimension

A visual image provided by M. C. Escher is extremely useful in aiding the intuition here: his drawing *Dragon* (Fig. 76). Its most salient feature is, of course, its subject matter-a dragon biting its tail, with all the Gödelian connotations which that carries. But there is a deeper theme to this picture. Escher himself wrote the following most interesting comments. The first comment is about a set of his drawings all of which are concerned with "the conflict between the flat and the spatial"; the second comment is about Dragon in particular.

I Our three-dimensional space is the only true reality we know. The two-dimensional is every bit as fictitious as the four-dimensional, for nothing is flat, not even the most finely polished mirror. And yet we stick to the convention that a wall or a piece of paper is flat, and curiously enough, we still go on, as we have done since time immemorial, producing illusions of space on just such plane surfaces as these. Surely it is a bit absurd to draw a few lines and then claim: "This is a house". This odd situation is the theme of the next five pictures (Including Dragon)

II. However much this dragon tries to be spatial, he remains completely flat. Two incisions are made in the paper on which he is printed. Then it is folded in such a way as to leave two square openings. But this dragon is an obstinate beast, and in' spite of his two dimensions he persists in assuming that he has three; so he sticks his head through one of the holes and his tail through the others [5]

This second remark especially is a very telling remark. The message is that no matter how cleverly you try to simulate three dimensions in two, you are always missing some "essence of three-dimensionality". The dragon tries very hard to fight his two-dimensionality. He defies the two-dimensionality of the paper on which he thinks he is drawn, by sticking his head through it; and yet all the while, we outside the drawing can see the pathetic futility of it all, for the dragon and the holes and the folds are all merely two-dimensional simulations of those concepts, and not a one of them is real. But the dragon cannot step out of his two-dimensional space, and cannot

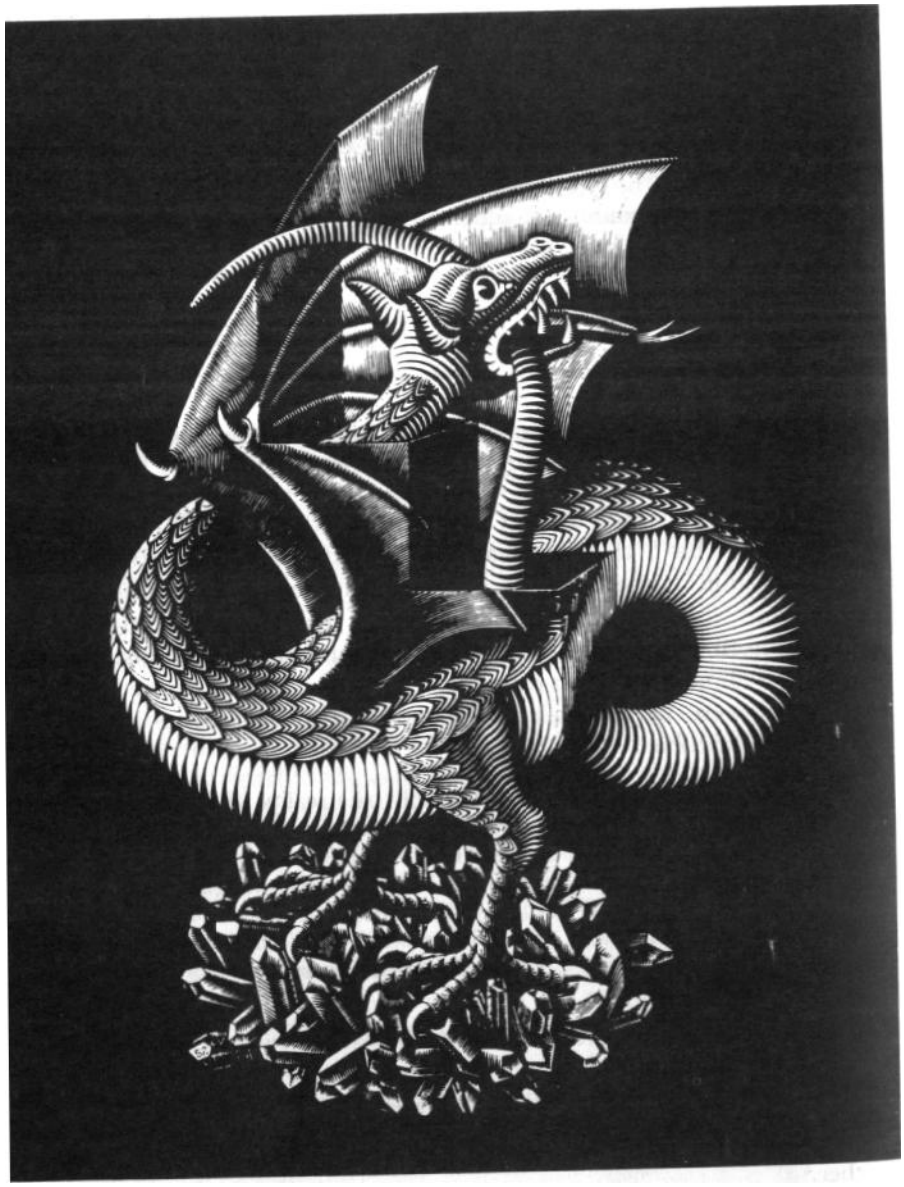


FIGURE 76. Dragon, *by M. C. Escher (wood-engraving*, 1952).

know it as we do. We could, in fact, carry the Escher picture any number of steps further. For instance, we could tear it out of the book, fold it, cut holes in it, pass it through itself, and photograph the whole mess, so that it again becomes two-dimensional. And to that photograph, we could once again do the same trick. Each time, at the instant that it becomes two- Matter how 'cleverly we seem to have simulated three dimensions inside two—it becomes vulnerable to being cut and folded again.

Now with this wonderful Escherian metaphor, let us return to the program versus the human. We were talking about trying to encapsulate the "Gödelizing operator" inside the program itself. Well, even if we had written a program which carried the operation out, that program would not capture the essence of Gödel's method. For once again, we, outside the system, could still "zap" it in a way which it couldn't do. But then are we arguing with, or against, Lucas

The Limits of Intelligent Systems

Against. For the very fact that we cannot write a program to do "Gödelizing" must make us somewhat suspicious that we ourselves could do it in every case. It is one thing to make the argument in the abstract that Gödelizing "can be done"; it is another thing to know how to do it in every particular case. In fact, as the formal systems (or programs) escalate in complexity, our own ability to "Gödelize" will eventually begin to waver. It must, since, as we have said above, we do not have any algorithmic way of describing how to perform it. If we can't tell explicitly what is involved in applying the Gödel method in all cases, then for each of us there will eventually come some case so complicated that we simply can't figure out how to apply it.

Of course, this borderline of one's abilities will be somewhat ill-defined, just as is the borderline of weights which one can pick up off the ground. While on some days you may not be able to pick up a 250-pound object, on other days maybe you can. Nevertheless, there are no days whatsoever on which you can pick up a 250-ton object. And in this sense, though everyone's Godelization threshold is vague, for each person, there are systems which lie far beyond his ability to Godelize.

This notion is illustrated in the Birthday Cantatatata. At first, it seems obvious that the Tortoise can proceed as far as he wishes in pestering Achilles. But then Achilles tries to sum up all the answers in a single swoop. This is a move of a different character than any that has gone before, and is given the new name 'co'. The newness of the name is quite important. It is the first example where the old naming scheme-which only included names for all the natural numbers-had to be transcended. Then come some more extensions, some of whose names seem quite obvious, others of which are rather tricky. But eventually, we run out of names once again-at the point where the answer-schemas

$\Omega, \omega^{\omega}, \omega^{\omega\omega}$ …..

are all subsumed into one outrageously complex answer schema. The altogether new name 'e,,' is supplied for this one. And the reason a new name is needed is that some fundamentally new kind of step has been taken—a sort of irregularity has been encountered. Thus a new name must be applied ad hoc.

There Is No Recursive Rule for Naming Ordinals.

Now offhand you might think that these irregularities in the progression >m ordinal to ordinal (as these names of infinity are called) could be handled by a computer program. That is, there would be a program to produce new names in a regular way, and when it ran out of gas, it would invoke the "irregularity handler", which would supply a new name, and pass control back to the simple one. But this will not work. It turns out that irregularities themselves happen in irregular ways, and one would need o a second-order program-that is, a program which makes new programs which make new names. And even this is not enough. Eventually, a third-order program becomes necessary. And so on, and so on.

All of this perhaps ridiculous-seeming complexity stems from a deep °theorem, due to Alonzo Church and Stephen C. Kleene, about the structure of these "infinite ordinals", which says:

> There is no recursively related notation-system which gives a name to every constructive ordinal.

hat "recursively related notation-systems" are, and what "constructive ordinals" are, we must leave to the more technical sources, such as Hartley)gets' book, to explain. But the intuitive idea has been presented. As the ordinals get bigger and bigger, there are irregularities, and irregularities in e irregularities, and irregularities in the irregularities in the irregularities, etc. No single scheme, no matter how complex, can name all e ordinals. And from this, it follows that no algorithmic method can tell w to apply the method of Gödel to all possible kinds of formal systems. ad unless one is rather mystically inclined, therefore one must conclude at any human being simply will reach the limits of his own ability to 5delize at some point. From there on out, formal systems of that complex, though admittedly incomplete for the Gödel reason, will have as much power as that human being.

Other Refutations of Lucas

Now this is only one way to argue against Lucas' position. There are others, possibly more powerful, which we shall present later. But this counterargument has special interest because it brings up the fascinating concept trying to create a computer program which can get outside of itself, see itself completely from the outside, and apply the Gödel zapping-trick to itself. Of course this is just as impossible as for a record player to be able to ay records which would cause it to break.

But-one should not consider **TNT** defective for that reason. If there a defect anywhere, it is not in **TNT**, but in our expectations of what it should he able to do. Furthermore, it is helpful to realize that we are equally vulnerable to the word trick which Gödel transplanted into mathematical formalisms: the Epimenides paradox. This was quite cleverly pointed out

by C. H. Whitely, when he proposed the sentence "Lucas cannot consistently assert this sentence." If you think about it, you will see that (1) it is true, and yet (2) Lucas cannot consistently assert it. So Lucas is also "incomplete" with respect to truths about the world. The way in which he mirrors the world in his brain structures prevents him from simultaneously being "consistent" and asserting that true sentence. But Lucas is no more vulnerable than any of us. He is just on a par with a sophisticated formal system.

An amusing way to see the incorrectness of Lucas' argument is to translate it into a battle between men and women ... In his wanderings, Loocus the Thinker one day comes across an unknown object-a woman. Such a thing he has never seen before, and at first he is wondrous thrilled at her likeness to himself: but then, slightly scared of her as well, he cries to all the men about him, "Behold! I can look upon her face, which is something she cannot do-therefore women can never be like me!" And thus he proves man's superiority over women, much to his relief, and that of his male companions. Incidentally, the same argument proves that Loocus is superior to all other males, as well-but he doesn't point that out to them. The woman argues back: "Yes, you can see my face, which is something I can't do-but I can see your face, which is something you can't do! We're even." However, Loocus comes up with an unexpected counter: "I'm sorry, you're deluded if you think you can see my face. What you women do is not the same as what we men do-it is, as I have already pointed out, of an inferior caliber, and does not deserve to be called by the same name. You may call it `womanseeing'. Now the fact that you can 'womansee' my face is of no import, because the situation is not symmetric. You see?" "I womansee," womanreplies the woman, and womanwalks away .. .

Well, this is the kind of "heads-in-the-sand" argument which you have to be willing to stomach if you are bent on seeing men and women running ahead of computers in these intellectual battles.

Self-Transcendence-A Modern Myth

It is still of great interest to ponder whether we humans ever can jump out of ourselves-or whether computer programs can jump out of themselves. Certainly it is possible for a program to modify itself-but such modifiability has to be inherent in the program to start with, so that cannot be counted as an example of "jumping out of the system". No matter how a program twists and turns to get out of itself, it is still following the rules inherent in itself. It is no more possible for it to escape than it is for a human being to decide voluntarily not to obey the laws of physics. Physics is an overriding system, from which there can be no escape. However, there is a lesser ambition which it is possible to achieve: that is, one can certainly Jump from a subsystem of one's brain into a wider subsystem. One can step out of ruts on occasion. This is still due to the interaction of various subsystems of one’s brain, but it can feel very much like stepping entirely out of oneself. Similarly, it is entirely conceivable that a partial ability to “step outside of itself” could be embodied in a computer program.

However, it is important to see the distinction between perceiving oneself, and transcending oneself. You can gain visions of yourself in all sorts of rays-in a mirror, in photos or movies, on tape, through the descriptions if others, by getting psychoanalyzed, and so on. But you cannot quite break out of your own skin and be on the outside of yourself (modern occult movements, pop psychology fads, etc. notwithstanding). **TNT** can talk about itself, but it cannot jump out of itself. A computer program can modify itself but it cannot violate its own instructions-it can at best change some parts of itself by obeying its own instructions. This is reminiscent of the numerous paradoxical question, "Can God make a stone so heavy that he can't lift it?"

Advertisement and Framing Devices

[his drive to jump out of the system is a pervasive one, and lies behind all progress in art, music, and other human endeavors. It also lies behind such trivial undertakings as the making of radio and television commercials. [his insidious trend has been beautifully perceived and described by Irving Goffman in his book *Frame Analysis*:

> For example, an obviously professional actor completes a commercial pitch and, with the camera still on him, turns in obvious relief from his task, now to take real pleasure in consuming the product he had been advertising.
>
> This is, of course, but one example of the way in which TV and radio commercials are coming to exploit framing devices to give an appearance of naturalness that (it is hoped) will override the reserve auditors have developed. Thus, use is currently being made of children's voices, presumably because these seem unschooled; street noises, and other effects to give the impression of interviews with unpaid respondents; false starts, filled pauses, byplays, and overlapping speech to simulate actual conversation; and, following Welles, the interception of a firm's jingle commercials to give news of its new product, alternating occasionally with interception by a public interest spot, this presumably keeping the faith of the auditor alive.
>
> The more that auditors withdraw to minor expressive details as a test of genuineness, the more that advertisers chase after them. What results is a sort of interaction pollution, a disorder that is also spread by the public relations consultants of political figures, and, more modestly, by micro-sociology.'

Here we have yet another example of an escalating "**TC**-battle"-the antagonists this time being Truth and Commercials.

Simplicio, Salviati, Sagredo: Why Three?

There is a fascinating connection between the problem of jumping out of ie system and the quest for complete objectivity. When I read Jauch's four dialogues in Are Quanta Real? based on Galileo's four Dialogues Concerning Two New Sciences, I found myself

wondering why there were three characters participating. Simplico, Salviati and Sagredo. Why wouldn't two have

sufficed: Simplicio, the educated simpleton, and Salviati, the knowledgeable thinker? What function does Sagredo have? Well, he is supposed to be a sort of neutral third party, dispassionately weighing the two sides and coming out with a "fair" and "impartial" judgment. It sounds very balanced, and yet there is a problem: Sagredo is always agreeing with Salviati, not with Simplicio. How come Objectivity Personified is playing favorites? One answer, of course, is that Salviati is enunciating correct views, so Sagredo has no choice. But what, then, of fairness or "equal time"?

By adding Sagredo, Galileo (and Jauch) stacked the deck *more* against Simplicio, rather than less. Perhaps there should be added a yet higher level Sagredo-someone who will be objective about this whole situation ... You can see where it is going. We are getting into a never-ending series of "escalations in objectivity", which have the curious property of never getting any more objective than at the first level: where Salviati is simply right, and Simplicio wrong. So the puzzle remains: why add Sagredo at all? And the answer is, it gives the illusion of stepping out of the system, in some intuitively appealing sense.

Zen and "Stepping Out"

In Zen, too, we can see this preoccupation with the concept of transcending the system. For instance, the koan in which Tozan tells his monks that "the higher Buddhism is not Buddha". Perhaps, self-transcendence is even the central theme of Zen. A Zen person is always trying to understand more deeply what he is, by stepping more and more out of what he sees himself to be, by breaking every rule and convention which he perceives himself to be chained by-needless to say, including those of Zen itself. Somewhere along this elusive path may come enlightenment. In any case (as I see it), the hope is that by gradually deepening one's self-awareness, by gradually widening the scope of "the system", one will in the end come to a feeling of being at one with the entire universe.

Edifying Thoughts of a Tobacco Smoker

Achilles has been invited to the Crab's home.

Achilles: I see you have made a few additions since I was last here, Mr. Crab. Your new paintings are especially striking.

Crab: Thank you. I am quite fond of certain painters-especially Rene Magritte. Most of the paintings I have are by him. He's my favorite artist.

Achilles: They are very intriguing images, I must say. In some ways, these paintings by Magritte remind me of works by MY favorite artist, M. C. Escher.

Crab: I can see that. Both Magritte and Escher use great realism in exploring the worlds of paradox and illusion; both have a sure sense for the evocative power of certain visual symbols, and-something which even their admirers often fail to point out-both of them have a sense of the graceful line.

Achilles: Nevertheless, there is something quite different about them. I wonder how one could characterize that difference.

Crab: It would be fascinating to compare the two in detail.

Achilles: I must say, Magritte's command of realism is astonishing. For instance, I was quite taken in by that painting over there of a tree with a giant pipe behind it.

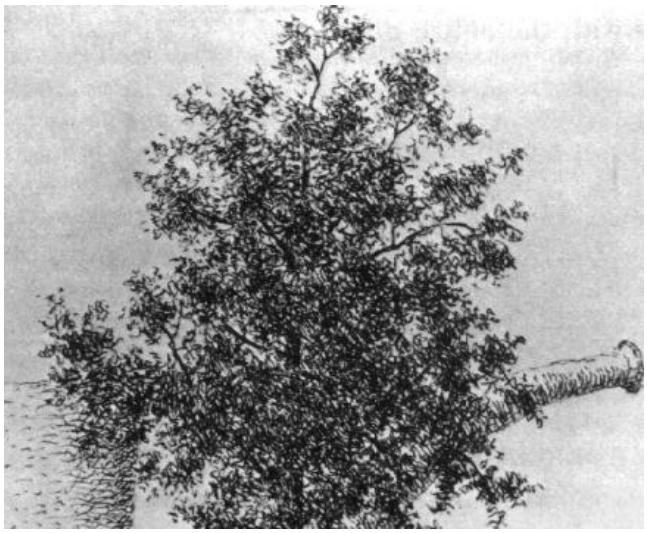


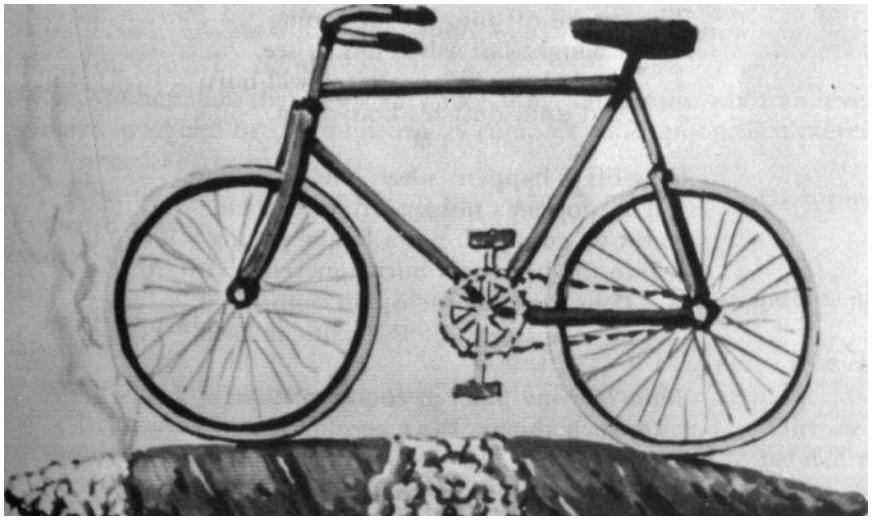
FIGURE 77. The Shadows, by Rene Magritte (1966).

Crab: You mean a normal pipe with a tiny tree in front of it!
Achilles: Oh, is that what it is? Well, in any case, when I first spotted it, I was convinced I was smelling pipe smoke! Can you imagine how silly I felt?
Crab: I quite understand. My guests are often taken in by that one.

(So saying, he reaches up, removes the pipe from behind the tree in the painting, turns over and taps it against the table, and the room begins to reek of pipe tobacco. He begins packing in a new wad of tobacco.)

This is a fine old pipe, Achilles. Believe it or not, the bowl has a copper lining, which makes it age wonderfully.
Achilles: A copper lining! You don't say!
Crab (pulls out a box of matches, and lights his pipe): Would you care for a smoke, Achilles?
Achilles: No, thank you. I only smoke cigars now and then.
Crab: No problem! I have one right here! (Reaches out towards another Magritte painting, featuring a bicycle mounted upon a lit cigar.)
Achilles: Uhh-no thank you, not now.
Crab: As you will. I myself am an incurable tobacco smoker. Which reminds me-you undoubtedly know of Old Bach's predilection for pipe smoking?
Achilles: I don't recall exactly.
Crab: Old Bach was fond of versifying, philosophizing, pipe smoking, and

FIGURE 78. State of Grace, by Rene Magritte (1959).



music making (not necessarily in that order). He combined all four into a droll poem which he set to music. It can be found in the famous musical notebook he kept for his wife, Anna Magdalena, and it is called

Edifying Thoughts of a Tobacco Smoker'

Whene'er I take my pipe and stuff it
And smoke to pass the time away,
My thoughts, as I sit there and puff it,
Dwell on a picture sad and gray:
It teaches me that very like
Am I myself unto my pipe.

Like me, this pipe so fragrant burning
Is made of naught but earth and clay;
To earth I too shall be returning.
It falls and, ere I'd think to say,
It breaks in two before my eyes;
In store for me a like fate lies.

No stain the pipe's hue yet cloth darken;
It remains white. Thus do I know
That when to death's call I must harken
My body, too, all pale will grow.
To black beneath the sod 'twill turn,
Likewise the pipe, if oft it burn.

Or when the pipe is fairly glowing,
Behold then, instantaneously,
The smoke off into thin air going,
Till naught but ash is left to see.
Man's fame likewise away will burn
And unto dust his body turn.

How oft it happens when one's smoking:
The stopper's missing from its shelf,
And one goes with one's finger poking
Into the bowl and burns oneself.
If in the pipe such pain cloth dwell,
How hot must be the pains of hell.

Thus o'er my pipe, in contemplation
Of such things, I can constantly
Indulge in fruitful meditation,
And so, puffing contentedly,
On land, on sea, at home, abroad
I smoke my pipe and worship God.

A charming philosophy, is it not?

Achilles: Indeed. Old Bach was a turner of phrases quite pleasin'.

Crab: You took the very words from my mouth. You know, in my time I have tried to write clever verses. But I fear mine don't measure up to much. I don't have such a way with words.

Achilles: Oh, come now, Mr. Crab. You have-how to put it?-quite a penchant for trick'ry and teasin'. I'd be honored if you'd sing me one of your songs, Mr. C.

Crab: I'm most flattered. How about if I play you a record of myself singing one of my efforts? I don't remember when it dates from. Its title is "A Song Without Time or Season".

Achilles: How poetic!

(The Crab pulls a record from his shelves, and walks over to a huge, complex piece of apparatus. He opens it up, and inserts the record into an ominous-looking mechanical mouth. Suddenly a bright flash of greenish light sweeps over the surface of the record, and after a moment, the record is silently whisked into some hidden belly of the fantastic machine. A moment passes, and then the strains of the Crab's voice ring out.)

A turner of phrases quite pleasin',
Had a penchant for trick'ry and teasin'.
In his songs, the last line
Might seem sans design;
What I mean is, without why or wherefore.

Achilles: Lovely! Only, I'm puzzled by one thing. It seems to me your song, the last line is

Crab: Sans design?

Achilles: No ... What I mean is, without rhyme or reason. Crab: You could be right.

Achilles: Other than that, it's a very nice song, but I must say I am even more intrigued by this monstrously complex contraption. Is it merely an oversized record player?

Crab: Oh, no, it's much more than that. This is my Tortoise-chomping record player. Achilles: Good grief!

Crab: Well, I don't mean that it chomps up Tortoises. But it chomps up records produced by Mr. Tortoise.

Achilles: Whew! That's a little milder. Is this part of that weird musical battle that evolved between you and Mr. T some time ago?

Crab: In a way. Let me explain a little more fully. You see, Mr. Tortoise's sophistication had reached the point where he seemed to be able to destroy almost any record player I would obtain.

Achilles: But when I heard about your rivalry, it seemed to me you had at last come into possession of an invincible phonograph—one with a

built-in TV camera, minicomputer and so on, which could take itself apart and rebuild itself in such a way that it would not be destroyed.

Crab: Alack and alas! My plan was foiled. For Mr. Tortoise took advantage of one small detail which I had overlooked: the subunit which directed the disassembly and reassembly processes was itself stable during the entire process. That is, for obvious reasons, it could not take itself apart and rebuild itself, so it stayed intact.

Achilles: Yes, but what consequences did that have.

Crab: Oh, the direst ones! For you see, Mr. T focused his method down onto that subunit entirely.

Achilles: How is that=

Crab: He simply made a record which would induce fatal vibrations in the one structure he knew would never change-the disassembly reassembly subunit.

Achilles: Oh, I see ... Very sneaky.

Crab: Yes, so I thought, too. And his strategy worked. Not the first time, mind you. I thought I had outwitted him when my phonograph survived his first onslaught. I laughed gleefully. But the next time, he returned with a steely glint in his eye, and I knew he meant business. I placed his new record on my turntable. Then, both of us eagerly watched the computer-directed subunit carefully scan the grooves. then dismount the record, disassemble the record player, reassemble it in an astonishingly different way, remount the record-and then slowly lower the needle into the outermost groove.

Achilles: Golly!

Crab: No sooner had the first strains of sound issued forth than a loud SMASH! filled the room. The whole thing fell apart, but particularly badly destroyed was the assembler-disassembler. In that painful instant I finally realized, to my chagrin, that the Tortoise would ALWAYS be able to focus down upon-if you'll pardon the phrase-the Achilles' heel of the system.

Achilles: Upon my soul! You must have felt devastated.

Crab: Yes, I felt rather forlorn for a while. But, happily, that was not the end of the story. There is a sequel to the tale, which taught me a valuable lesson, which I may pass on to you. On the Tortoise's recommendation, I was browsing through a curious book filled with strange Dialogues about many subjects, including molecular biology, fugues, Zen Buddhism, and heaven knows what else.

Achilles: Probably some crackpot wrote it. What is the book called:'

Crab: If I recall correctly, it was called Copper, Silver, Gold: an Indestructible Metallic Alloy.

Achilles: Oh, Mr. Tortoise told me about it, too. It's by a friend of his, who, it appears, is quite taken with metal-logic.

Crab- I wonder which friend it is ... Anyway_ in one of the Dialogues, I encountered some Edifying Thoughts on the Tobacco Mosaic Virus, ribosomes, and other strange things I had never heard of.

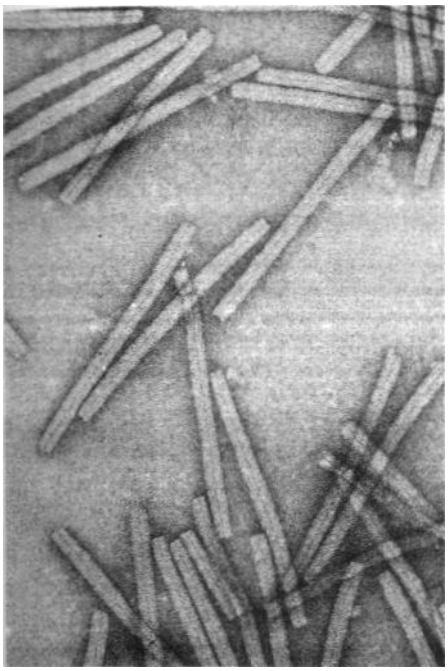


FIGURE 79. Tobacco Mosaic Virus.
From A. Lehninger, Biochemistry (New York: Worth Publishers, 1976).

Achilles: What is the Tobacco Mosaic Virus? What are ribosomes?

Crab: I can't quite say, for I'm a total dunce when it comes to biology. All I know is what I gathered from that Dialogue. There, it said that Tobacco Mosaic Viruses are tiny cigarette-like objects that cause a disease in tobacco plants.

Achilles: Cancer?

Crab: No, not exactly, but

Achilles: What next? A tobacco plant smoking and getting cancer! Serves it right!

Crab: I believe you've jumped to a hasty conclusion, Achilles. Tobacco plants don't SMOKE these "cigarettes". The nasty little "cigarettes" just come and attack them, uninvited.

Achilles: I see. Well, now that I know all about Tobacco Mosaic Viruses, tell me what a ribosome is.

Crab: Ribosomes are apparently some sort of sub cellular entities which take a message in one form and convert it into a message in another form.

Achilles: Something like a teeny tape recorder or phonograph?

Crab: Metaphorically, I suppose so. Now the thing which caught my eye was a line where this one exceedingly droll character mentions the fact that ribosomes-as well as Tobacco Mosaic Viruses and certain other bizarre biological structures-possess "the baffling ability to spontaneously self-assemble, Those were his exact words.

Achilles: That was one of his droller lines, I take it.

Crab: That's just what the other character in the Dialogue thought. But that's a preposterous interpretation of the statement. (*The Crab draws deeply from his pipe, and puffs several billows of smoke into the air.)*

Achilles: Well, what does "spontaneous self-assembly" mean, then?

Crab: The idea is that when some biological units inside a cell are taken apart, they can spontaneously reassemble themselves-without being directed by any other unit. The pieces just come together, and presto!-they stick.

Achilles: That sounds like magic. Wouldn't it be wonderful if a full-sized record player could have that property? I mean, if a miniature "record player" such as a ribosome can do it, why not a big one? That would allow you to create an indestructible phonograph, right? Any time it was broken, it would just put itself together again.

Crab: Exactly my thought. I breathlessly rushed a letter off to my manufacturer explaining the concept of self-assembly, and asked him if he could build me a record player which could take itself apart and spontaneously self-assemble in another form.

Achilles: A hefty bill to fill.

Crab: True; but after several months, he wrote to me that he had succeeded, at long last-and indeed he sent me quite a hefty bill. One fine day, ho! My Grand Self-assembling Record Player arrived in the mail, and it was with great confidence that I telephoned Mr. Tortoise, and invited him over for the purpose of testing my ultimate record player.

Achilles: So this magnificent object before us must be the very machine of which you speak.

Crab: I'm afraid not, Achilles.

Achilles: Don't tell me that once again ...

Crab: What you suspect, my dear friend is unfortunately the case. I don't pretend to understand the reasons why. The whole thing is too painful to recount. To see all those springs and wires chaotically strewn about on the floor, and puffs of smoke here and there-oh, me ...

Achilles: There, there, Mr. Crab, don't take it too badly.

Crab: I'm quite all right; I just have these spells every so often. Well, to go on, after Mr. Tortoise's initial gloating, he at last realized how sorrowful I was feeling, and took pity. He tried to comfort me by explaining that it couldn't be helped-it all had to do with somebody-or-other's "Theorem", but I couldn't follow a word of it. It sounded like "Turtle's Theorem".

Achilles: I wonder if it was that "Gödel's Theorem" which he spoke of once before to me ... It has a rather sinister ring to it. Crab: It could be. I don't recall.

Achilles: I can assure you, Mr. Crab, that I have followed this tale with the utmost empathy for your position. It is truly sad. But, you mentioned that there was a silver lining. Pray tell, what was that?

Crab: Oh, yes—the silver lining. Well eventually, I abandoned my quest after "Perfection" in phonographs, and decided that I might do better

to tighten up my defenses against the Tortoise's records. I concluded that a more modest aim than a record player which can play anything is simply a record player that can SURVIVE: one that will avoid getting destroyed-even if that means that it can only play a few particular records.

Achilles: So you decided you would develop sophisticated anti-Tortoise mechanisms at the sacrifice of being able to reproduce every possible sound, eh?

Crab: Well ... I wouldn't exactly say I "decided" it. More accurate would be to say that I was FORCED into that position.

Achilles: Yes, I can see what you mean.

Crab: My new idea was to prevent all "alien" records from being played on my phonograph. I knew my own records are harmless, and so if I prevented anyone else from infiltrating THEIR records, that would protect my record player, and still allow me to enjoy my recorded music.

Achilles: An excellent strategy for your new goal. Now does this giant thing before us represent your accomplishments to date along those lines?

Crab: That it does. Mr. Tortoise, of course, has realized that he must change HIS strategy, as well. His main goal is now to devise a record which can slip past my censors-a new type of challenge.

Achilles: For your part, how are you planning to keep his and other "alien" records out?

Crab: You promise you won't reveal my strategy to Mr. T, now?

Achilles: Tortoise's honor.

Crab: What!?

Achilles: Oh-it's just a phrase I've picked up from Mr. T. Don't worry-I swear your secret will remain secret with me.

Crab: All right, then. My basic plan is to use a LABELING technique. To each and every one of my records will be attached a secret label. Now the phonograph before you contains, as did its predecessors, a television camera for scanning the records, and a computer for processing the data obtained in the scan and controlling subsequent operations. My idea is simply to chomp all records which do not bear the proper label!

Achilles: Ah, sweet revenge! But it seems to me that your plan will be easy to foil. All Mr. T needs to do is to get a hold of one of your records, and copy its label!

Crab: Not so simple, Achilles. What makes you think he will be able to tell the label from the rest of the record? It may be better integrated than you suspect.

Achilles: Do you mean that it could be mixed up somehow with the actual music?

Crab: Precisely. But there is a way to disentangle the two. It requires sucking the data off the record visually and then--

Achilles: Is that what that bright green Hash was for?

Crab: That's right. That was the TV camera scanning the grooves. The groove-patterns were sent to the minicomputer, which analyzed the musical style of the piece I had put on-all in silence. Nothing had been played yet.

Achilles: Then is there a screening process, which eliminates pieces which aren't in the proper styles?

Crab: You've got it, Achilles. The only records which can pass this second test are records of pieces in my own style-and it will be hopelessly difficult for Mr. T to imitate that. So you see, I am convinced I will win this new musical battle. However, I should mention that Mr. T is equally convinced that somehow, he will manage to slip a record past my censors.

Achilles: And smash your marvelous machine to smithereens?

Crab: Oh, no-he has proved his point on that. Now he just wants to prove to me that he can slip a record-an innocuous one-by me, no matter what measures I take to prevent it. He keeps on muttering things about songs with strange titles, such as "I Can Be Played on Record Player X". But he can't scare MtE! The only thing that worries me a little is that, as before, he seems to have some murky arguments which ... which ... (He trails off into silence. Then, looking quite pensive, he takes a few puffs on his pipe.)

Achilles: Hmm ... I'd say Mr. Tortoise has an impossible task on his hands. He's met his match, at long last!

Crab: Curious that you should think so ... I don't suppose that you know Henkin's Theorem forwards and backwards, do you?

Achilles: Know WHOSE Theorem forwards and backwards? I've never heard of anything that sounds like that. I'm sure it's fascinating, but I'd rather hear more about "music to infiltrate phonographs by". It's an amusing little story. Actually, I guess I can fill in the end. Obviously, Mr. T will find out that there is no point in going on, and so he will sheepishly admit defeat, and that will be that. Isn't that exactly it?

Crab: That's what I'm hoping, at least. Would you like to see a little bit of the inner workings of my defensive phonograph?

Achilles: Gladly. I've always wanted to see a working television camera.

Crab: No sooner said than done, my friend. (Reaches into the gaping"mouth" of the large phonograph, undoes a couple of snaps, and pulls out a neatly packaged instrument.) You see, the whole thing is built of independent modules, which can be detached and used independently. This TV camera, for instance, works very well by itself. Watch the screen over there, beneath the painting with the flaming tuba. (He points the camera at Achilles, whose face instantly appears on the large screen.)

Achilles: Terrific! May I try it out?

Crab: Certainly.

Achilles: (pointing the camera at the Crab. There YOU are, Mt Crab, on the screen.

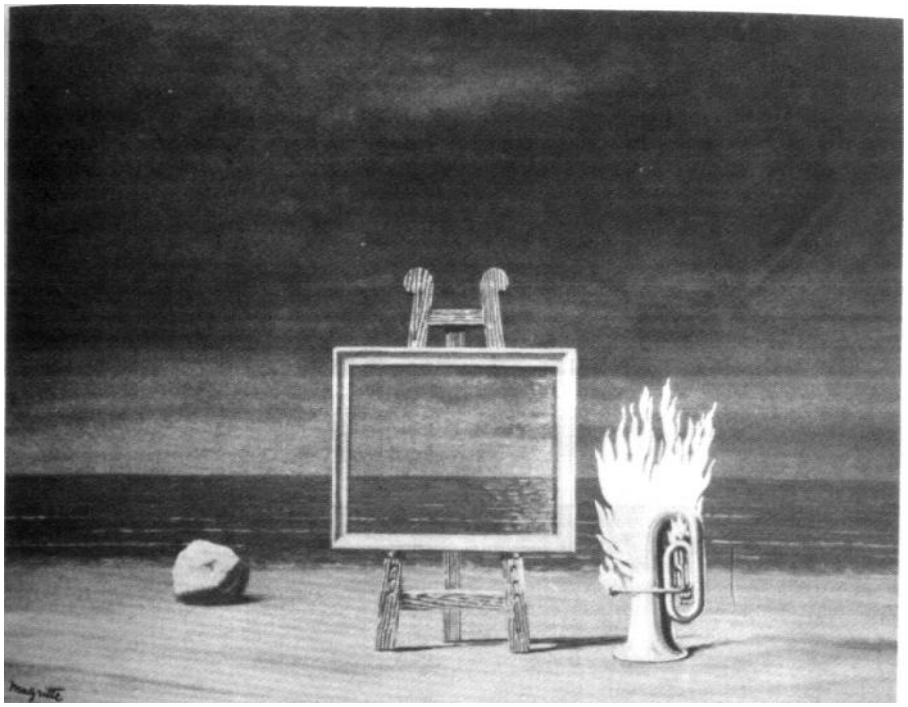


FIGURE 80. The Fair Captive, by Rene Magritte (1947).

Crab: So I am.

Achilles: Suppose I point the camera at the painting with the burning tuba. Now it is on the screen, too!

Crab: The camera can zoom in and out, Achilles. You ought to try it. Achilles: Fabulous! Let me just focus down onto the tip of those flames, where they meet the picture frame ... It's such a funny feeling to be able to instantaneously "copy" anything in the room-anything I want-onto that screen. I merely need to point the camera at it, and it pops like magic onto the screen.

Crab: ANYTHING in the room, Achilles? Achilles: Anything in sight, yes. That's obvious.

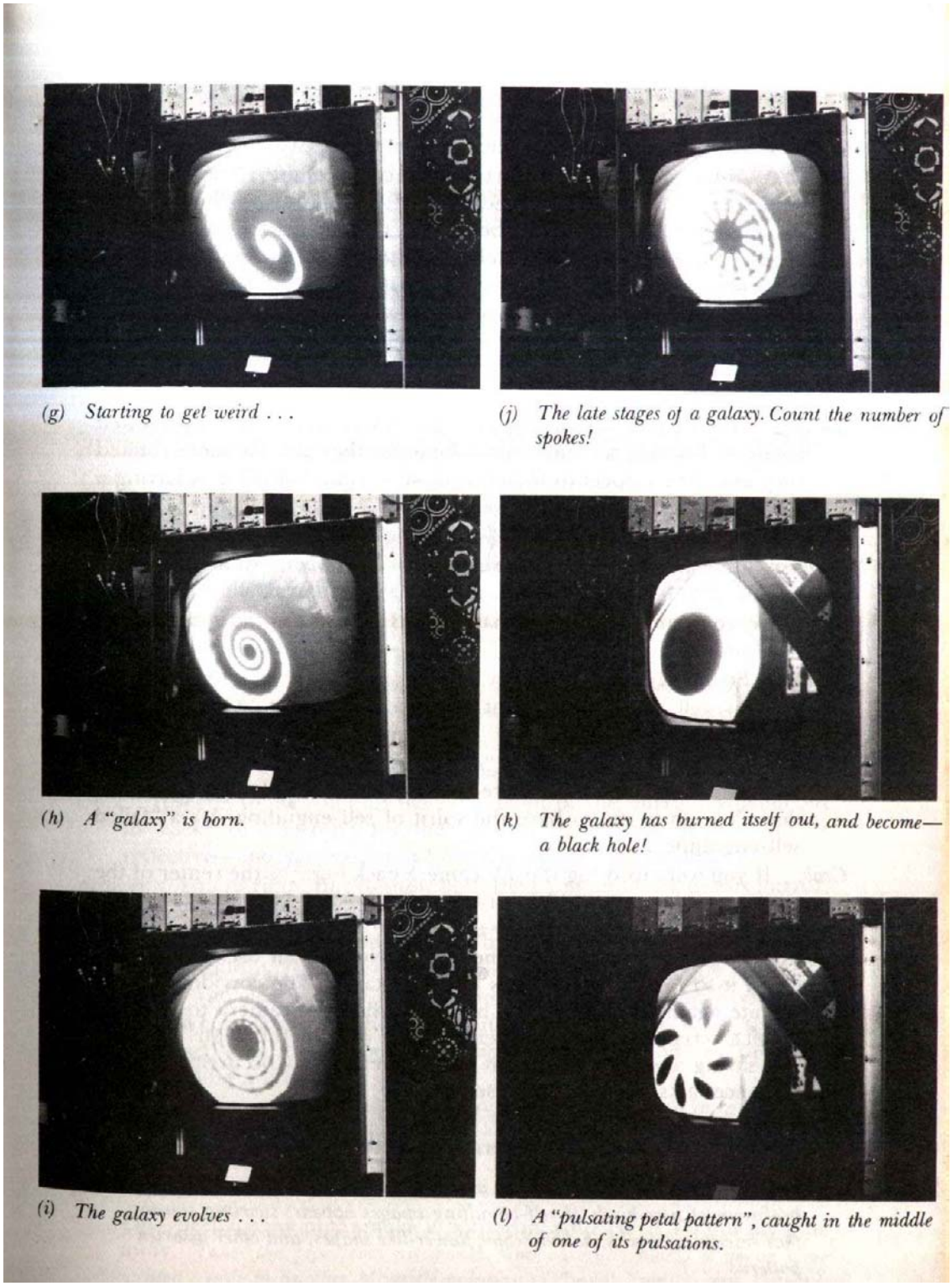
Crab: What happens, then, if you point the camera at the flames on the TV screen?

(Achilles shifts the camera so that it points directly at that part of the television screen on which the flames are-or were-displayed.)

Achilles: Hey, that's funny! That very act makes the flames DISAPPEAR from the screen! Where did they go?

Crab: You can't keep an image still on the screen and move the camera at the same time.

Achilles: So I see… But I don't understand what's on the screen now—not at all! It seems to be a strange long corridor. Yet I'm certainly not



(g) *Starting to get weird . . .*

(j) *The late stages of a galaxy. Count the number of spokes!*

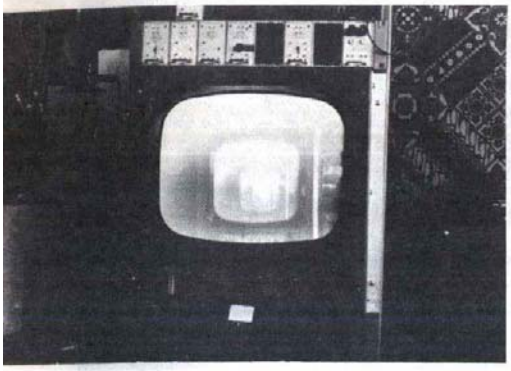
(h) *A "galaxy" is born.*

(k) *The galaxy has burned itself out, and become—a black hole!*

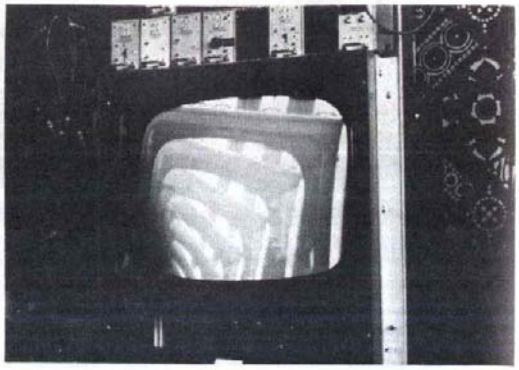
(i) *The galaxy evolves . . .*

(l) *A "pulsating petal pattern", caught in the middle of one of its pulsations.*

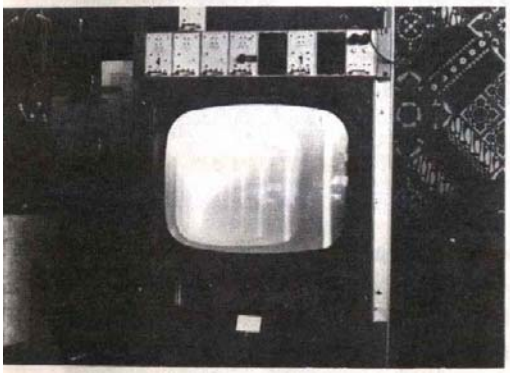
FIGURE 81. *Twelve self-engulfing TV screens. I would have included one more, had 13 not been prime*



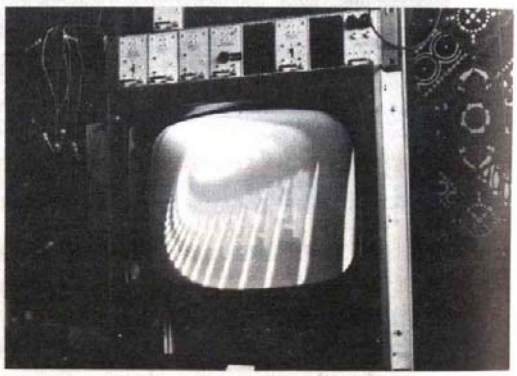
) *The simplest case.*



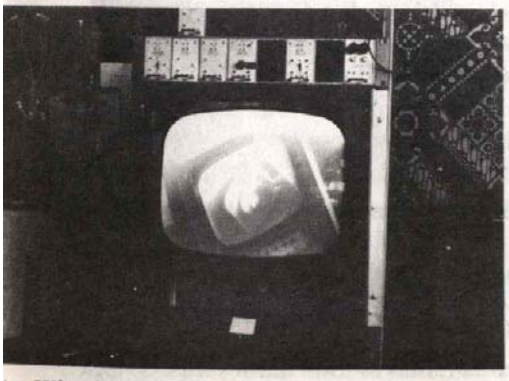
(d) A "failed self-engulfing".



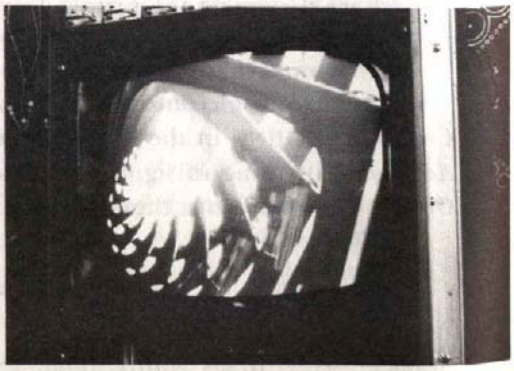
) *Achilles' "corridor".*



(e) What happens when you zoom in.



) *What happens when you rotate the camera.*



(f) Combined effect of rotation and zooming.

pointing the camera down any corridor. I'm merely pointing it at an ordinary TV screen.

Crab: Look more carefully, Achilles. Do you really see a corridor?

Achilles: Ahhh, now I see. It's a set of nested copies of the TV screen itself, getting smaller and smaller and smaller ... Of course! The image of the flames HAD to go away, because it came from my- pointing the camera at the PAINTING. When I point the camera at the SCREEN, then the screen itself appears, with whatever is on the screen at the time which is the screen itself, with whatever is on the screen at the time which is the screen itself, with

Crab: I believe I can fill in the rest, Achilles. Why- don't you try rotating the camera?

Achilles: Oh! I get a beautiful spiraling corridor! Each screen is rotated inside its framing screen, so that the littler they get, the more rotated they are, with respect. to the outermost screen. This idea of having a TV screen "engulf itself" is weird.

Crab: What do you mean by "self-engulfing", Achilles?

Achilles: I mean, when I point the camera at the screen-or at part of the screen. THAT'S self-engulfing.

Crab: Do you mind if I pursue that a little further? I'm intrigued by this new notion.

Achilles: So am I.

Crab: Very well, then. If you point the camera at a CORNER of the screen, is that still what you mean by "self-engulfing"?

Achilles: Let me try it. Hmm-the "corridor" of screens seems to go off the edge, so there isn't an infinite nesting any more. It's pretty, but it doesn't seem to me to have the spirit of self-engulfing. It's a "failed self-engulfing".

Crab: If you were to swing the TV camera back towards the center of the screen, maybe you could fix it up again ...

Achilles (slowly and cautiously turning the camera): Yes! The corridor is getting longer and longer ... There it is! Now it's all back. I can look down it so far that it vanishes in the distance. The corridor became infinite again precisely at the moment when the camera took in the WHOLE screen. Hmm-that reminds me of something Mr. Tortoise was saying a while back, about self-reference only occurring when a sentence talks about ALL of itself ...

Crab: Pardon me?

Achilles: Oh, nothing just muttering to myself.

(As Achilles plays with the lens and other controls on the camera, a profusion of new kinds of self-engulfing images appear: swirling spirals that resemble galaxies, kaleidoscopic flower-like shapes, and other assorted patterns ...)

Crab: You seem to be having a grand time.

Achilles: (turns away from the camera); I'll say! What a wealth of images this simple idea can produce! (He glances back at the screen, and a look of

astonishment crosses his face.) Good grief, Mr. Crab! There's a pulsating petal-pattern on the screen! Where do the pulsations come from? The TV is still, and so is the camera.

Crab: You can occasionally set up patterns which change in time. This is because there is a slight delay in the circuitry between the moment the camera "sees" something, and the moment it appears on the screen around a hundredth of a second. So if you have a nesting of depth fifty or so, roughly a half-second delay will result. If somehow a moving image gets onto the screen-for example, by you putting your finger in front of the camera-then it takes a while for the more deeply nested screens to "find out" about it. This delay then reverberates through the whole system, like a visual echo. And if things are set up so the echo doesn't die away, then you can get pulsating patterns.

Achilles: Amazing! Say-what if we tried to make a TOTAL self-engulfing?

Crab: What precisely do you mean by that?

Achilles: Well, it seems to me that this stuff with screens within screens is interesting, but I'd like to get a picture of the TV camera AND the screen, ON the screen. Only then would I really have made the system engulf itself. For the screen is only PART of the total system.

Crab: I see what you mean. Perhaps with this mirror, you can achieve the effect you want.

(The Crab hands him a mirror, and Achilles maneuvers the mirror and camera in such a way that the camera and the screen are both pictured on the screen.)

Achilles: There! I've created a TOTAL self-engulfing!

Crab: It seems to me you only have the front of the mirror-what about its back? If it weren't for the back of the mirror, it wouldn't be reflective-and you wouldn't have the camera in the picture.

Achilles: You're right. But to show both the front and back of this mirror, I need a second mirror.

Crab: But then you'll need to show the back of that mirror, too. And what about including the back of the television, as well as its front? And then there's the electric cord, and the inside of the television, and

Achilles: Whoa, whoa! My head's beginning to spin! I can see that this "total self-engulfing project" is going to pose a wee bit of a problem. I'm feeling a little dizzy.

Crab: I know exactly how you feel. Why don't you sit down here and take your mind off all this self-engulfing? Relax! Look at my paintings, and you'll calm down.

(Achilles lies down, and sighs.)

Oh-perhaps my pipe smoke is bothering you? Here, I'll put my pipe away. (*Takes the pipe from his mouth, and carefully places it above some written words in another .Magritte painting.*) There! Feeling any better?

Achilles: I’m still a little woozy, (Points at the Magritte.) That’s an interesting painting. I like the way it’s framed, especially the shiny inlay inside the wooden frame.

FIGURE 82. The Air and the Song, by Rene Magritte (1964).

Crab: Thank you. I had it specially done-it's a gold lining.

Achilles: A gold lining? What next? What are those words below the pipe? They aren't in English, are they?

Crab: No, they are in French. They say, "Ceci n'est pas une pipe." That means, "This is not a pipe". Which is perfectly true. 4chilles: But it is a pipe! You were just smoking it!

Crab: Oh, you misunderstand the phrase, I believe. The word "ceci" refers to the painting, not to the pipe. Of course the pipe is a pipe. But a painting is not a pipe.

Achilles: I wonder if that "ceci" inside the painting refers to the WHOLE painting, or just to the pipe inside the painting. Oh, my gracious! That would be ANOTHER self-engulfing! I'm not feeling at all well, Mr. Crab. I think I'm going to be sick ...